

Healthcare Capacity Monitoring and Operational Analytics Using Time-Series Data: An Exploratory Data Analysis Approach

Abstract

Healthcare systems require continuous monitoring to ensure that patient demand is effectively balanced with available resources. This study presents a data-driven analytical framework for monitoring healthcare capacity using operational data collected from the United States Department of Health and Human Services (HHS) and Customs and Border Protection (CBP). Daily time-series records from 2023 to 2025 were analyzed to evaluate system load, patient movement, capacity pressure, and operational sustainability.

The analysis involved data preprocessing, quality validation, feature engineering, exploratory data analysis (EDA), temporal trend analysis, and healthcare pressure identification. Key metrics including Total System Load, Net Daily Intake, Care Load Growth Rate, and Backlog Indicators were derived to quantify operational performance. A Streamlit-based interactive dashboard was developed to enable real-time visualization of healthcare capacity and facilitate evidence-based decision-making.

The proposed framework demonstrates how healthcare administrators and policymakers can leverage analytical tools to identify periods of capacity strain, optimize staffing and resource allocation, and improve humanitarian response planning.

Keywords: Healthcare Analytics, Time Series Analysis, Exploratory Data Analysis, Streamlit Dashboard, Capacity Planning, Humanitarian Operations.

1. Introduction

Healthcare systems experience continuous fluctuations in patient volume, making efficient resource management essential for maintaining service quality. Daily operational data generated by HHS and CBP contains valuable information regarding patient intake, transfers, and discharges. However, without an integrated analytical framework, it becomes difficult to monitor operational efficiency or identify capacity constraints.

This project develops a comprehensive healthcare analytics framework capable of transforming raw operational data into actionable insights through statistical analysis, visualization, and interactive reporting.

2. Problem Statement

Although daily operational data is collected, HHS lacks a centralized analytical framework capable of continuously assessing:

- Total healthcare system load.
- Balance between patient inflow and outflow.
- Capacity stress and relief periods.
- Sustainability of healthcare delivery over time.

Consequently, healthcare administrators have limited visibility into operational bottlenecks and resource utilization.

3. Research Objectives

Primary Objectives

- Quantify daily and cumulative healthcare system load.
- Measure operational balance between CBP and HHS.
- Identify periods of capacity stress.
- Analyze patient intake, transfers, and discharges.

Secondary Objectives

- Support healthcare staffing decisions.
 - Improve situational awareness for policymakers.
 - Enable evidence-based humanitarian response evaluation.
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4. Dataset Description

The dataset consists of daily operational records collected between 2023 and 2025.

The primary variables include:

- Date
- Children in CBP Custody
- Children in HHS Care
- Children Transferred from CBP
- Children Discharged from HHS Care

The dataset was transformed into a continuous time-series format for longitudinal analysis.

5. Methodology

The analytical workflow consists of the following stages:

Data Ingestion

- Import CSV dataset.
- Convert Date into datetime format.
- Sort records chronologically.
- Create a continuous daily index.

Data Quality Assessment

- Identify missing dates.
- Detect duplicate observations.
- Validate operational constraints.
- Flag reporting anomalies.

Feature Engineering

The following healthcare metrics were derived:

- Total System Load
- Net Daily Intake
- Care Load Growth Rate
- Backlog Indicator
- Rolling Averages (7-Day and 14-Day)
- Prolonged Capacity Strain

Exploratory Data Analysis

The EDA included:

- Descriptive statistics
- Time-series visualization
- Weekly and monthly aggregation
- Trend decomposition
- Distribution analysis
- Rolling average analysis

6. Exploratory Data Analysis (EDA)

The exploratory analysis revealed several important operational characteristics:

- Daily healthcare demand fluctuated significantly across the observation period.

- Weekly aggregation reduced short-term variability while revealing consistent demand patterns.
- Monthly analysis demonstrated seasonal changes in healthcare system utilization.
- Rolling averages successfully highlighted sustained periods of increased operational pressure.
- Capacity stress was concentrated during prolonged periods of positive net intake.

The derived metrics enabled effective monitoring of healthcare workload and patient movement throughout the study period.

7. Key Insights

The analysis generated the following operational insights:

- Total healthcare system load exhibited clear temporal variation.
 - Positive Net Daily Intake corresponded with increasing healthcare backlog.
 - Seven-day and fourteen-day rolling averages effectively identified persistent capacity pressure.
 - Several prolonged strain periods indicated sustained demand exceeding normal operational levels.
 - Variability analysis demonstrated periods of unstable healthcare demand requiring adaptive resource allocation.
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8. Dashboard Development

A web-based dashboard was developed using Streamlit to provide interactive operational monitoring.

Dashboard modules include:

- KPI Summary Cards
- System Load Overview
- CBP vs HHS Comparison
- Net Intake Trend
- Rolling Average Analysis
- Pressure Identification
- Downloadable Filtered Reports

Interactive capabilities include:

- Date range filtering
 - Daily, weekly, and monthly views
 - Metric selection
 - Dynamic visualizations
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9. Recommendations

Based on the analytical findings, the following recommendations are proposed:

1. Establish continuous monitoring of healthcare system load using automated dashboards.
 2. Introduce predictive analytics to forecast future capacity requirements.
 3. Increase staffing during prolonged high-load periods identified through rolling averages.
 4. Develop early-warning indicators using Net Daily Intake and Backlog metrics.
 5. Integrate operational analytics into routine healthcare planning to improve responsiveness and resource allocation.
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10. Conclusion

This research demonstrates the effectiveness of time-series analytics for monitoring healthcare capacity and operational performance. By integrating exploratory data analysis, feature engineering, and interactive visualization, the proposed framework transforms daily operational records into actionable intelligence.

The developed Streamlit dashboard provides healthcare administrators with a practical decision-support tool for monitoring capacity trends, identifying operational stress, and supporting evidence-based planning. The framework is scalable and can be extended to predictive modeling and real-time healthcare monitoring applications.

Future Work

Future enhancements may include:

- Machine learning-based patient load forecasting.
 - Anomaly detection using statistical learning techniques.
 - Real-time dashboard integration with live healthcare databases.
 - Geographic visualization of healthcare capacity.
 - Automated alert generation during capacity stress events.
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